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# **Customer Behavioral Segmentation for Targeted Marketing**

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**Project Id: AIML-10**

Name : **TIRTH SHARMA** (Team Leader)

Class Roll Number: **73**

University Roll Number : **2028884**

Section and Group: **Q (G2)**

Name : **RITIK RAJ**

Class Roll Number: **62**

University Roll Number : **2028823**

Section and Group: **Q (G2)**

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## 1. Problem Description

- The Customer Segmentation System is a machine learning-based project developed to group customers based on their purchasing behavior. The main objective of this project is to identify different types of customers such as budget customers and premium customers using clustering techniques.
- In today's business environment, understanding customer behavior is very important for decision-making. This system helps businesses analyze customer data and divide them into meaningful groups based on their spending patterns and purchase frequency.
- The project uses an unsupervised learning approach, specifically the K-Means clustering algorithm, which groups similar data points without predefined labels. Features such as Annual Spending and Purchase Frequency are used to identify patterns among customers.
- The system also includes data preprocessing, feature scaling, model selection, evaluation, and visualization techniques to ensure accurate clustering. This helps businesses in targeted marketing, customer retention, and improving overall performance.

## 2.Architecture



## **2. Models Used:-**

### **K-Means Clustering**

K-Means is an unsupervised machine learning algorithm used to divide data into K clusters. It works by assigning each data point to the nearest centroid and updating centroids iteratively.

#### **Working of K-Means:**

- Select number of clusters (K)
- Initialize centroids
- Assign data points to nearest centroid
- Update centroids
- Repeat until convergence

#### **In this project:**

- K was selected using Elbow Method
- Customers are divided into:
  - Budget Customers
  - Premium Customers

#### **Advantages:**

- Simple and efficient
- Works well for numerical data
- Easy to understand and implement

### 3. Evaluation metrics used

#### 1. WCSS (Within Cluster Sum of Squares)

WCSS measures how compact the clusters are. It calculates the sum of squared distances between each point and its cluster centroid.

- Lower WCSS → better clustering
  - Used in Elbow Method
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#### ◆ 2. Silhouette Score

Silhouette Score measures how well-separated the clusters are.

Range:

- **+1 → Excellent clustering**
- **0 → Overlapping clusters**
- **Negative → Poor clustering**

It considers:

- Intra-cluster distance (within cluster)
  - Inter-cluster distance (between clusters)
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#### ◆ 3. Visual Evaluation

Scatter plots are used to visually check cluster separation.

- Clear separation → good clustering
- Overlapping → poor clustering

## 4. Preliminary Results and Visualizations

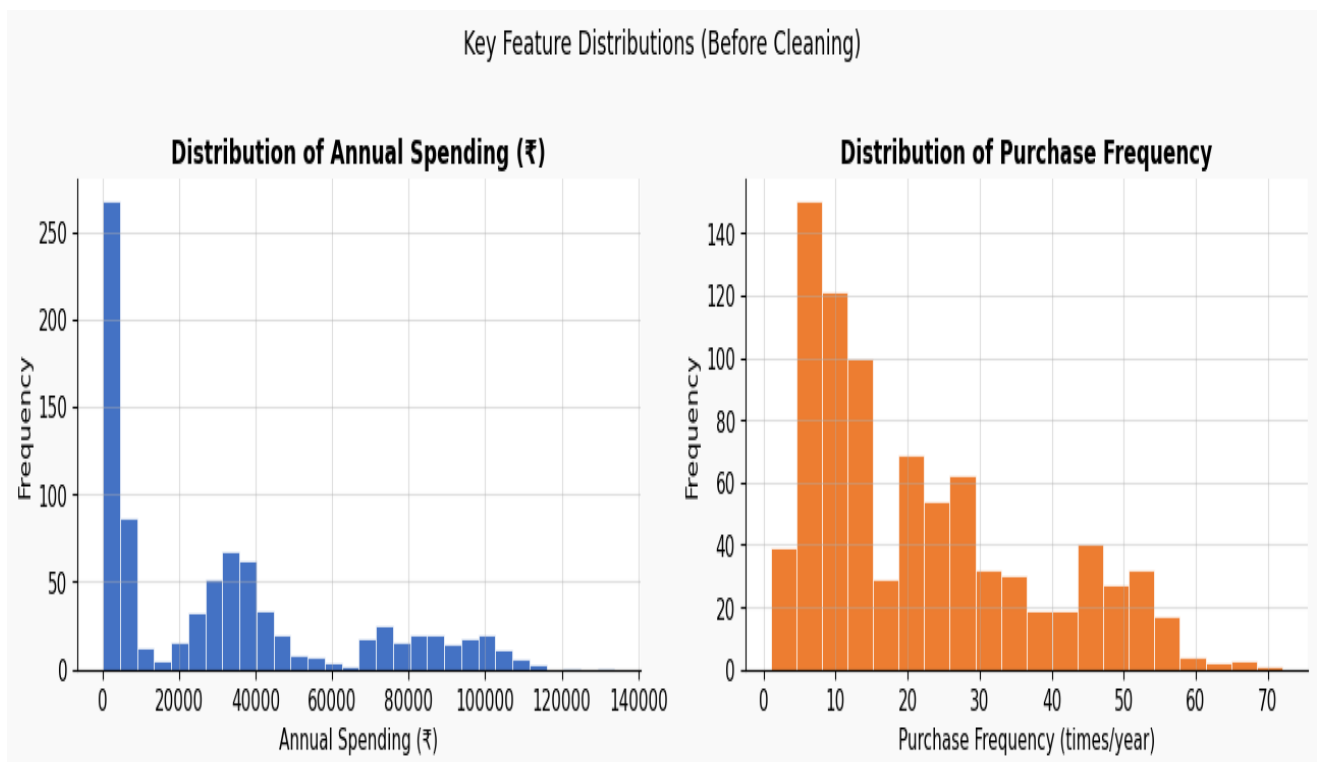
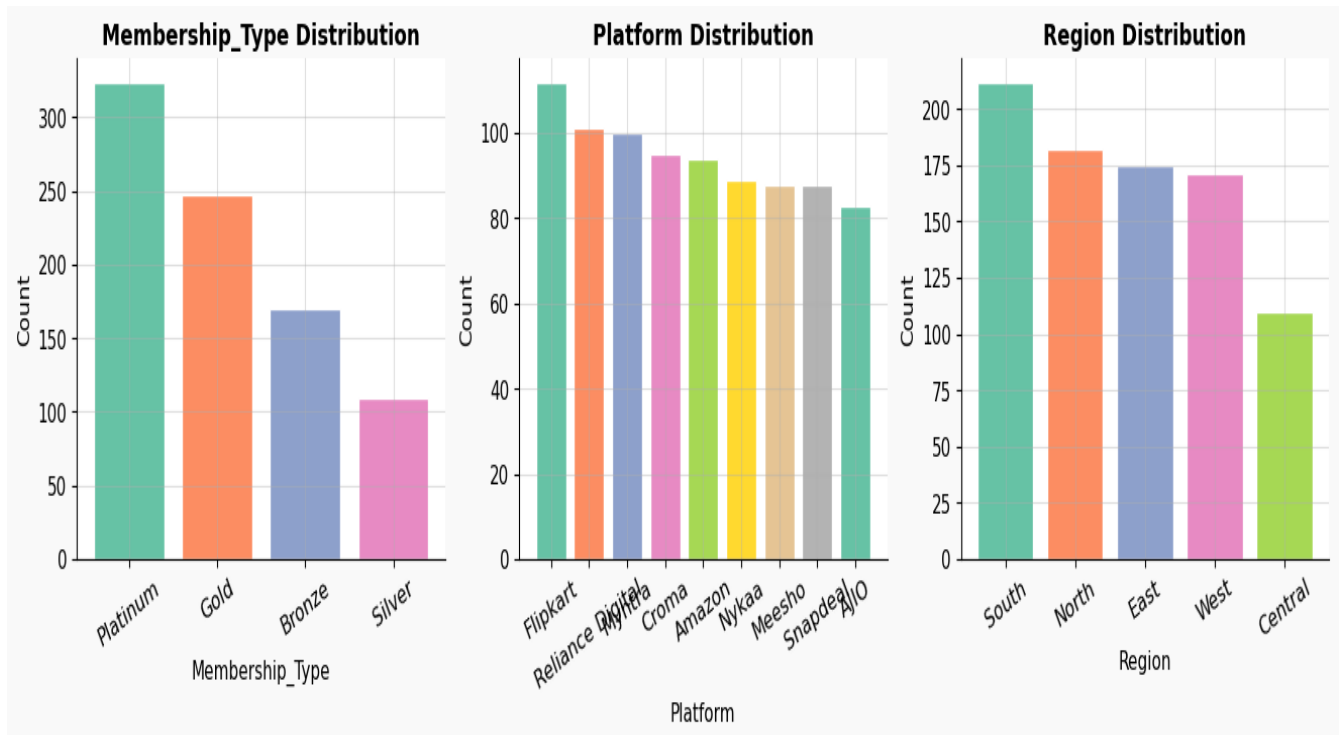


Fig.1 Features Distribution

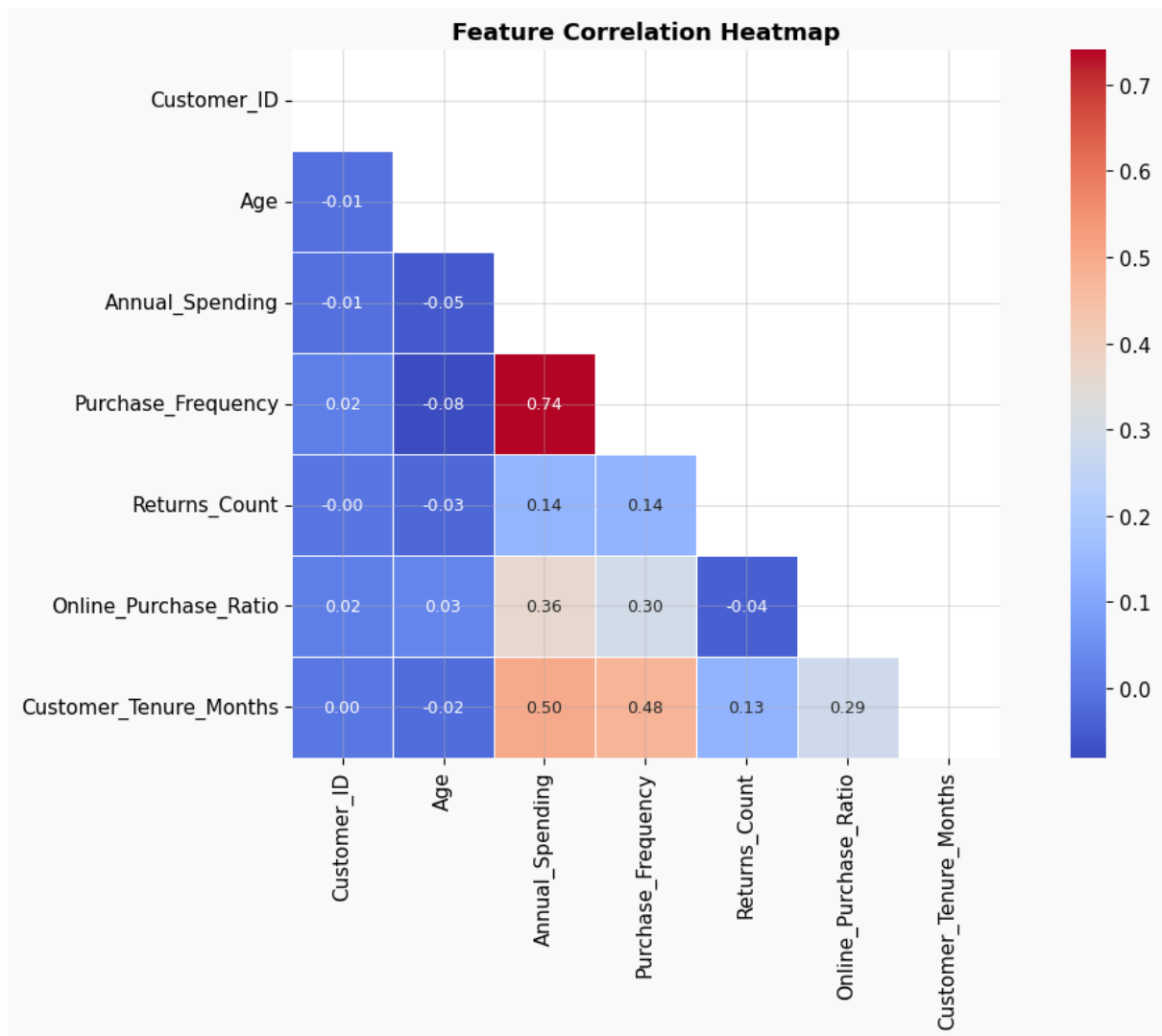


Fig.2 Heatmap showing correlation.



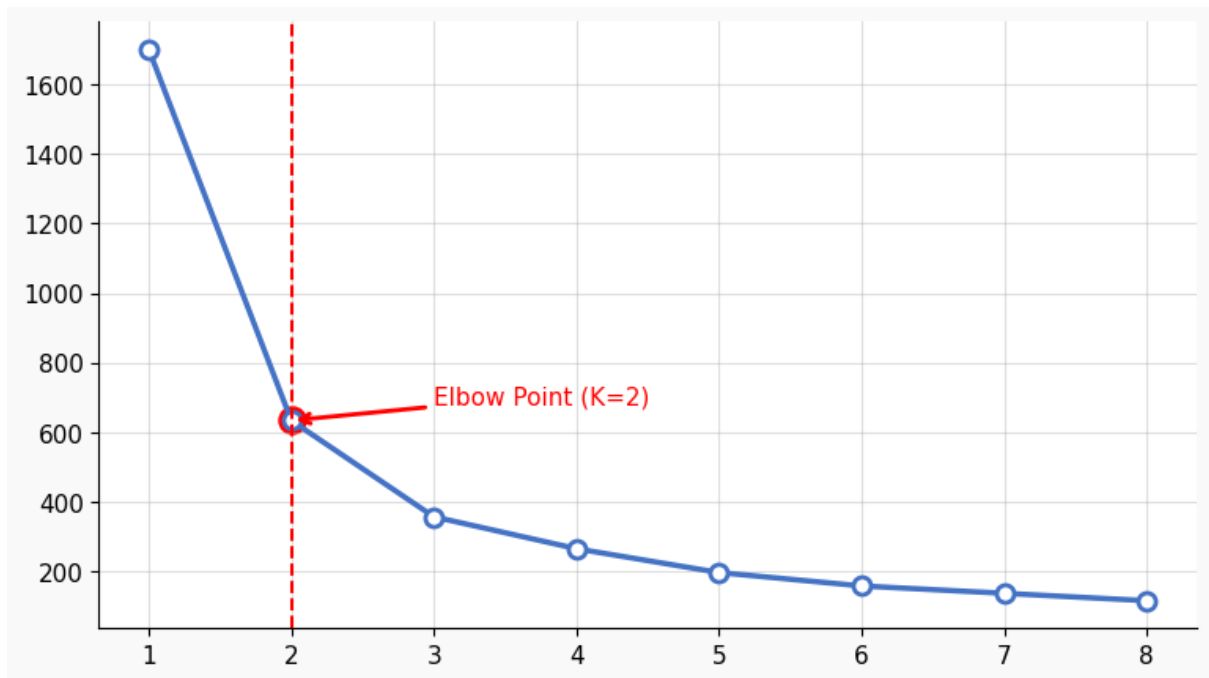


Fig.3 Elbow Method (optimal k)

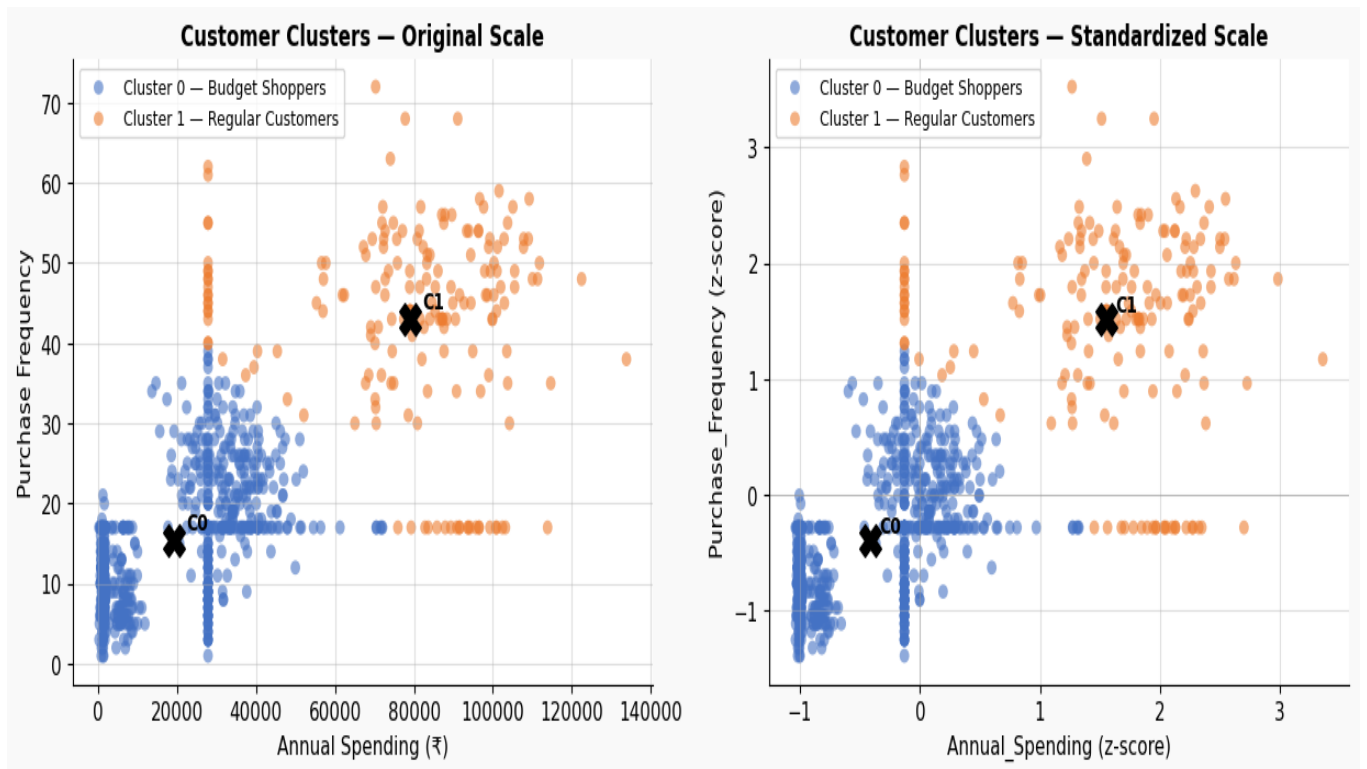


Fig.4 clusters(0 and 1)

## 6. References:

- **Platform Required:**

- **Jupyter Notebook** – For interactive coding, visualization, and documentation.
- **Python 3.x** - The core programming language for the implementation.

- **Books and Link Sources:**

- *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* by Aurélien Géron.
- Campus x Youtube channel-[https://youtu.be/Zja1aZXrlDE?si=a8Y\\_8AGZcda5uXUc](https://youtu.be/Zja1aZXrlDE?si=a8Y_8AGZcda5uXUc)
- Scikit-learn Official Documentation: `sklearn.cluster.KMeans`.
- Kaggle Tutorial: "Customer Segmentation using K-Means Clustering".